

§ 1

Fields

§1.1 Terminology and properties

Def 1.1 A field is a set with two binary operations, called addition

$$+ : F \times F \rightarrow F$$

and multiplication

$$\cdot : F \times F \rightarrow F$$

such that the following properties hold

A1 (Commutativity)

$$\forall a, b \in F, \quad a + b = b + a$$

A2 (Associativity)

$$\forall a, b, c \in F, \quad a + (b + c) = (a + b) + c$$

A3 (Existence of zero)

$$\exists \text{ element } 0 \in F \text{ s.t. } \forall a \in F,$$

$$0 + a = a + 0 = a$$

A4 (Existence of negatives)

$$\forall a \in F, \exists -a \in F \text{ st.}$$

$$a + (-a) = 0$$

M1 (Commutativity)

$$\forall a, b \in F, \quad a \cdot b = b \cdot a$$

M2 (Associativity)

$$\forall a, b, c \in F, \quad a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

M3 (Existence of identity)

$$\exists \text{ element } 1 \in F \text{ st. } \forall a \in F$$

$$1 \cdot a = a \cdot 1 = a$$

M4 (Existence of multiplicative inverses)

$$\forall a \in F \text{ with } a \neq 0, \exists a^{-1} \in F \text{ st.}$$

$$a \cdot a^{-1} = a^{-1} \cdot a = 1$$

D (Distributive property)

$$\forall a, b, c \in F, \quad a \cdot (b + c) = a \cdot b + a \cdot c$$

Examples of fields

not
yet
defined

$\mathbb{R}$: usual addition and multiplication

$\mathbb{Q}$: usual addition and multiplication

(Non-example) $\mathbb{Z}$ because there is
no multiplicative inverse for 2.

(Advanced example)

Let $S = \{\bar{0}, \bar{1}\}$, a set of two elements.

Define

+	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{1}$
$\bar{1}$	$\bar{1}$	$\bar{0}$

$\times$	$\bar{0}$	$\bar{1}$
$\bar{0}$	$\bar{0}$	$\bar{0}$
$\bar{1}$	$\bar{0}$	$\bar{1}$

Check S is a field.

Remark 1.2

- The (additive) 0 is unique.

Pf: Let 0 and $0'$ be zeros in A .

Then $0 + 0' = 0$ by (A3) using zeroness of 0

$0 + 0' = 0'$ by (A3) using zeroness of $0'$

$$\Rightarrow 0 = 0'$$

- For $a \in F$, the negative $-a$ is unique

Pf: Let x be an inverse of a .

$$x + a = 0 \Rightarrow (x + a) + (-a) = 0 + (-a)$$

(A3)

$$\Rightarrow (x + a) + (-a) = -a$$

(A2)

$$\Rightarrow x + (a + (-a)) = -a$$

(A4)

$$\Rightarrow x + 0 = -a$$

(A3)

$$\Rightarrow x = -a$$

- The (multiplicative) identity is unique
- The multiplicative inverse is unique

Remark 1.3 (Subtraction & division).

For $a, b \in F$, write $a - b$ for

$$a + (-b)$$

For $a, b \in F$ with $b \neq 0$, write $\frac{a}{b}$ for

$$a \cdot b^{-1}$$

Prop. 1.5 Let F be a field.

(a) If z and a are in F , with

$$z + a = a$$

then $z = 0$

$$\text{Pf: } (z + a) + (-a) = a + (-a).$$

$$z + (a + (-a)) = a + (-a) \quad A2$$

$$z + 0 = 0 \quad A4$$

$$z = 0 \quad A3$$

(b) If $u \in F$ and $b \in F$ with $b \neq 0$ and $ub = b$, then $u = 1$.

Pf: $(ub) \cdot (b^{-1}) = b \cdot b^{-1}$

$$u(b \cdot b^{-1}) = b \cdot b^{-1} \quad M2$$

$$u(1) = 1 \quad M4$$

$$u = 1 \quad M3$$

(c) If $a \in F$, then $0 \cdot a = 0$

Pf: $(0+0) \cdot a = 0 \cdot a$

(exercise) $0 \cdot a + 0 \cdot a = 0 \cdot a \quad (D)$

$$0 \cdot a = 0 \quad \text{by (a)}$$

(d) If $a \in F$, then $-(-a) = a$.

Pf $a + (-a) = (-a) + a = 0$

Uniqueness of additive inverse

$$\Rightarrow -(-a) = a$$

(e) If $a, b \in F$, then $(-a) \cdot b = -(ab) = a(-b)$

Pf: Check $(-a) \cdot b = -(ab)$

$$(-a)(b) + ab = (-a + a) \cdot b \quad D$$

$$= 0 \cdot b \quad A4$$

$$= 0 \quad \text{by (c)}$$

By uniqueness of $-(ab)$, $(-a)(b) = -(ab)$.

(f) If $a, b \in F$, then $(-a)(-b) = ab$

$$\text{Pf: } (-a)(-b) = -((-a)b) \quad (e)$$

$$= -(-(ab)) \quad (e)$$

$$= ab \quad (d)$$

§1.2 Ordered field

Def 1.6 A field is said to be ordered if there exists an ordering structure $\leq$ satisfying :

(01) For $a, b \in F$,

$$a \leq b \quad \text{or} \quad b \leq a$$

(02) For $a, b \in F$, if $a \leq b$ and $b \leq a$,
then $a = b$

(03) For $a, b, c \in F$, if $a \leq b$ and $b \leq c$,
then $a \leq b \leq c$.

(04) For $a, b, c \in F$, if $a \leq b$, then
 $a + c \leq b + c$

(05) For $a, b, c \in F$, if $a \leq b$ and $c \geq 0$,
then $ac \leq bc$

Ex 1.7 Suppose F is an ordered field.

(a) If $a \neq 0$, then either $0 < a$ or $0 < -a$

Pf: This follows from (O1) that

$$a < 0 \quad \text{or} \quad 0 < a$$

If we are in the latter case ($0 < a$),

$$\text{then } 0 + (-a) < a + (-a)$$

$$\Rightarrow -a < 0 \quad (\text{by (A3), (A4)})$$

(b) For any $a \in F$ and $a \neq 0$, $a^2 > 0$

Pf: By (a), $0 < a$ or $0 < -a$

$$\text{If } 0 < a, \text{ then } 0 < a^2 \quad (\text{by O5})$$

$$\text{If } 0 < -a, \text{ then } 0 < (-a)^2 \quad (\text{by O5})$$

$$\Rightarrow 0 < a^2 \quad (\text{by Prop 1.5 f})$$

(c) $0 < 1$

$$\text{Pf: } 1^2 = 1$$

$$\text{By (b), } 0 < 1$$

(d) If $0 < a$, then $0 < a^{-1}$

Pf: Suppose $0 < -a^{-1}$, By (O5)

$$0 \leq a \cdot (-a^{-1}) \Rightarrow 0 \leq -1$$

This contradicts (c). By (a), $0 < a$.

(e) For any $a, b \in F$, if $a < b$, then $-b < -a$

$$a < b \Rightarrow a + (-a - b) < b + (-a - b) \quad (\text{by O4})$$

$$\Rightarrow -b < -a$$

(f) For any $a, b \in F$, if $a < b$ and $c > 0$,
then $ac < bc$

Pf: By (05), $ac \leq bc$. If $ac = bc$, $(a-b) \cdot c = 0$
Since $a \neq b$, $a-b \neq 0$. So, $c = 0$. Contradiction.

(g) For any $a, b \in F$, if $0 < a$ and $0 < b$,
then $a < b \iff a^2 < b^2$

Pf: $a < b \xrightarrow{(f)} a^2 < ba \xrightarrow{(f)} a^2 < ba < b^2$

If $a = b$, then $a^2 = b^2$

If $b < a$, then $b^2 < a^2$

By (01), $a^2 < b^2 \implies a < b$.

(h) $1 + 1 \neq 0$.

Pf: By (c), $0 < 1$. By (04), $0 + 1 < 1 + 1$
 $1 < 1 + 1$

By (c), $0 < 1 < 1 + 1$

$1 + 1 \neq 0$

Examples 1.8

Def $\mathbb{N} = \{1, 2, 3, \dots\}$ (Peano's axiom)

- Add zeros and negatives

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

- Add inverses

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z}, q > 0 \right\}$$

- $\mathbb{Q}$ is an ordered field with $\leq$ given

$$\text{by } \frac{p}{q} \leq \frac{p'}{q'} \iff p'q - pq' > 0.$$

for $q, q' > 0$

Example 1.10 (Non-ordered field)

Define $\mathbb{Q}[i] =$ set of numbers of
the form $a + bi$

where i is a symbol called imaginary
number

$$\& \quad a, b \in \mathbb{Q}.$$

Addition: $(a_1 + b_1 i) + (a_2 + b_2 i) = (a_1 + a_2) + (b_1 + b_2) i$

Multiplication: $(a_1 + b_1 i)(a_2 + b_2 i)$

$$= (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1) i$$

$\mathbb{Q}[i]$ is a field, but not an ordered field

Zero : $0 + 0 i$

Identity : $1 + 0 i$

Additive inverse of $a + b i$:

$$(-a) + (-b) i$$

Multiplicative inverse of $a + b i$:

$$\frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i$$

$\mathbb{Q}[i]$ is not ordered because

$$i^2 = -1 < 0 \quad \text{by Ex 1.7(c)}$$

and so $\mathbb{Q}[i]$ does not satisfy Ex 1.7(b)

§ 2 Maximum & Minimum

§ 2.1

Def: Let S be a non-empty subset of an ordered field F .

(1) S is said to be bounded above if there exists $M \in F$ s.t.

$$\forall x \in S, \quad x \leq M$$

M is called an upper bound of S .

(2) S is said to be bounded below if there exists $m \in F$ s.t.

$$\forall x \in S, \quad x \geq m$$

m is called a lower bound of S

(3) S is said to be bounded if S is both bounded above and bounded below.

Example Let $S = \{x \in \mathbb{Q} : -1 < x \leq 2\}$.

S is bounded above by 2

S is bounded below by -1

Def Let S be a non-empty subset of F .

(1) The maximum of S is an element s_0

in S st. $x \leq s_0$ for all $x \in S$.

(2) The minimum of S is an element $\bar{s}_0$

in S st. $\bar{s}_0 \leq x$ for all $x \in S$.

Example. The set

$$\{x \in \mathbb{Q} : 0 \leq x < 8\}.$$

is bounded above, but the maximum does not exist.

- If a maximum/minimum of S exists, then it is unique.

Examples.

$$(1) \quad S_1 = \{x \in \mathbb{Q} : -1 < x \leq 2\}$$

$$\text{max.} = 2 \quad \text{min. not exist}$$

$$(2) \quad S_2 = \{x \in \mathbb{Q} : -2 < x \leq 1 \text{ or } 2 \leq x \leq 5\}$$

$$\text{max.} = 5 \quad \text{min. not exist}$$

$$(3) \quad S_3 = \{x \in \mathbb{Q} : -2 \leq x\}$$

$$\text{max. not exist} \quad \text{min} = -2$$

$$(4) \quad S_4 = \{x \in \mathbb{Q} : x^3 = 1\}.$$

$$\text{max} = 1 \quad \text{min} = 1$$

$$(5) \quad S_5 = \{x \in \mathbb{Q} : x^3 < 1\}$$

$$\text{max. not exist} \quad \text{min. not exist}$$

$$(6) \quad S_6 = \mathbb{N}$$

$$\text{max. not exist} \quad \text{min.} = 1$$

$$(7) \quad S_7 = \mathbb{Q}$$

$$\text{max. , min. not exist.}$$

§ Supremum and infimum

Def: Let F be an ordered field

Let $S \subseteq F$ and $S \neq \emptyset$.

(1) If S has an upper bound, then the supremum, denoted by $\sup S$, is the least upper bound of S in F .

$$\sup S = \min \left\{ u \in F : \begin{array}{l} u \text{ is an upper} \\ \text{bound of } S \end{array} \right\}$$

(2) If S has a lower bound, then the infimum, denoted by $\inf S$, is the greatest lower bound of S in F .

$$\inf S = \max \left\{ i \in F : \begin{array}{l} i \text{ is a lower} \\ \text{bound of } S \end{array} \right\}$$

Remark:

- $\sup(S)$ is not necessarily in S .

If $\sup(S) \in S$, then $\sup(S)$ is the maximum of S

- $\inf(S)$ is not necessarily in S .

If $\inf(S) \in S$, then $\inf(S)$ is the minimum of S .

- $\sup(S)$ is unique.

- $\inf(S)$ is unique.

Thm 3.3 (Criterion for finding supremum)

Let S be a non-empty subset of an ordered field F . Let u be an upper bound of S . Then

$$u = \sup(S) \iff \forall \varepsilon > 0, \exists s \in S \text{ s.t. } u - \varepsilon < s.$$

Proof:

($\Rightarrow$) Let $u = \sup(S)$.

Let $\varepsilon > 0$.

Then $u - \varepsilon$ is not an upper bound of S .

$\Rightarrow \exists s \in S$ s.t. $u - \varepsilon < s$.

($\Leftarrow$) Suppose u' is also an upper bound of S .

Let $\varepsilon = u - u'$.

If $\varepsilon > 0$, then

$$\exists x \in S \text{ s.t. } u - \varepsilon < x$$

$$\Rightarrow u' < x$$

$\Rightarrow$ this contradicts u' is an upper bound.

Examples

$$(1) \quad S_1 = \{x \in \mathbb{Q} : -1 < x \leq 2\}$$

$$\sup(S_1) = 2 \quad \inf(S_1) = -1$$

$$(2) \quad S_2 = \{x \in \mathbb{Q} : -2 < x \leq 1 \text{ or } 2 < x \leq 5\}$$

$$\sup(S_2) = 5 \quad \inf(S_2) = -2$$

$$(3) \quad S_3 = \{x \in \mathbb{Q} : -2 \leq x\}$$

$$\sup(S_3) \text{ not exist} \quad \inf(S_3) = -2$$

$$(4) \quad S_4 = \{x \in \mathbb{Q} : x^3 = 1\}.$$

$$\sup(S_4) = 1 \qquad \inf(S_4) = 1$$

$$(5) \quad S_5 = \{x \in \mathbb{Q} : x^3 < 1\}.$$

$$\sup(S_5) = 1 \qquad \inf(S_5) \text{ does not exist}$$

Ex 3.5 (ε -criterion for infimum version)

Let S be a non-empty subset of an ordered field F . Let l be a lower bound of S . Then

$$l = \inf(S) \iff \forall \varepsilon > 0, \exists s \in S \text{ s.t. } s > l + \varepsilon$$

Example Let $S = \{\frac{1}{n} : n \in \mathbb{N}\}$

$$\inf(S) = 0$$

Ex. 3.6

Let $S \subseteq$ ordered F . Suppose $\sup(S)$ exists. Let

$$-S = \{-x : x \in S\}.$$

Then $\inf(-S) = -\sup(S)$.

Pf: Step 1: $\inf(-S) \geq \sup(S)$.

Let $s = \sup(S)$.

For any $y \in -S$.

$$y = -x \text{ for some } x \in S.$$

$$\Rightarrow x = -y$$

$$\Rightarrow s \geq x = -y$$

$$\Rightarrow -s \leq y$$

$\therefore -s$ is a lower bound of $-S$.

Step 2: $\inf(-S) \leq \sup(S)$.

Suppose not. $\exists l > -s$ st.

$$l \leq y, \forall y \in -S$$

$$\Rightarrow l \leq -x, \forall x \in S$$

$$\Rightarrow -l \geq x, \forall x \in S$$

$$\Rightarrow s > -l \geq x, \forall x \in S$$

This contradicts that s is the least upper bound of S .

Ex. 3.7 Let S and T be two non-empty bounded above subsets of F .

$$S+T := \{ s+t \in F : s \in S, t \in T \}$$

Show that

$$\sup(S+T) = \sup(S) + \sup(T).$$

Pf: Step 1: Show that $\sup(S) + \sup(T)$ is an upper bound of $S+T$.

For any $x \in S+T$,

$$x = s + t \quad \text{for some } s \in S \text{ and } t \in T$$

$$s \leq \sup(S), \quad t \leq \sup(T)$$

$$\Rightarrow x \leq \sup(S) + \sup(T).$$

Step 2: Show $\sup(S) + \sup(T)$ is the least one

$$\text{By Thm 3.3, } \exists s' \in S \text{ s.t.} \\ \sup(S) - \frac{\varepsilon}{2} < s'$$

$$\text{Similarly, by Thm 3.3, } \exists t' \in T \text{ s.t.} \\ \sup(T) - \frac{\varepsilon}{2} < t'$$

$$\text{Thus, } \sup(S) + \sup(T) - \varepsilon < s' + t'$$

Since ε is arbitrary, Thm 3.3 implies that $\sup(S) + \sup(T)$ is $\sup(S+T)$.